EPS12V

This form factor introduced an additional 8-pin connector purely for the processor instead of the earlier 4-pin owing to higher power processors being released in the market. Though originally designed for servers with 2 sockets, this schema has been adopted for desktop use as well.

Connectors

The usual desktop SMPS comes with the following connectors

1. 20+4 pin motherboard connector for hooking up your motherboard.
2. 4+4 pin or simply one 4 pin connector for your CPU
3. 6 pin or 6+2 pin connector for your graphics card
4. 4 pin molex connector for your PATA / SATA drives and also for auxiliary uses like extra power and chassis fans.
5. 15 pin flat SATA power connectors

Efficiency

Efficiency of a power supply is defined as the ratio between the power pulled from the AC mains to the power provided by the power supply. Now, power supplies heat up a bit and we have a little loss in energy over there, similarly, there are other avenues for loss in energy. A good power supply has high efficiency and thus, is economical in the long run. A simple example is that if your power supply uses 100 W to provide 70 W as output then your SMPS efficiency is 70%. There is even a system of certification for SMPS. You'll see it as a small badge called 80+ efficiency. The following image should give you an idea of the different badges.

Parameters	Loading	80 Plus	Bronze	Silver	Gold
Efficiency	20%	80%	82%	85%	87%
	50%	80%	85%	88%	90%
	100%	80%	82%	85%	87%
Power Factor	50%	90% (@100% load)	90% (across the full range)		

Now the higher the efficiency the better the power supply, but the price increases likewise. This shouldn't be a concern as in the long run a SMPS with higher efficiency will save you a lot more on your electricity bill. Also since the efficiency is high the power supply does not lose much power to heat loss. Hence, it will be cooler in the long run too. It has been found that power supplies are the most efficient when the load on it is